Microbial Ecology and Bioremediation

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Microbial Ecology and Bioremediation (A06318)

Short Title:	Micro Ecology & Bioremediation		
Faculty	Science and Computing		
Department	Science		
Cluster	Biology		
Author	NKENNEDY		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

This module will give students a broad introduction to the major elements of microbial ecology and bioremediation. The interplay between microbe and macrobe will be studied, with an emphasis on the role of this interaction in areas such as the intestine and nutrient cycles in the environment. A series of laboratory-based experiments plus field trips will be used to reinforce the theory elements of this module.

Programmes

ENVI-0007	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	3 / 6 / M

Indicative Content

- Introduction to microbial ecology: examples of microbial ecosystems, microbe/microbe interactions, biofilms
- Microbe/macrobe interactions such as intestinal microflora, mycorrhizae
- Culture-dependent and independent analysis of microbial communities, e.g. plate counts, selective enrichment, microscope-based methods, Lux and GFP tagging, FISH, DNA-based profiling, including high throughput sequencing methods
- Wastewater treatment: traditional and newer remediation methods, nutrient removal from wastewater
- Bioremediation in soils and water: xenobiotics, pollutant bioremediation
- Anaerobic digestion and composting
- Field trips: field trips to facilities such as integrated wetland systems and wastewater treatment plants
- Laboratory practicals: analysis of microbial diversity in natural ecosystems, using both culture-dependent and independent techniques; isolation and characterisation of microbes from the environment

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Apply appropriate culture-dependent and -independent methodologies for analysis of microbial ecosystems.
2. Appraise the relationships microbes can have with each other and with macro-organisms.
3. Analyse the role of micro-organisms in a variety of microbial ecosystems, e.g. nutrient cycles, biodegradation of pollutants, animal and human intestines.
4. Express how bioremediation can help to provide solutions to current issues of concern to society.
5. Compare strategies for biological remediation and monitoring of pollution.
6. Prepare detailed records of activities of both laboratory and field visits.

Learning and Teaching Methods

- Lectures on the indicative content will be delivered, including active learning activities, peer-to-peer learning, and formative assessment (ungraded quizzes).
- Laboratory investigations plus field trips will reinforce and support theoretical elements of the module.
- Independent case study assignment on bioremediation will be carried out.

Assessment Methods

	Weighing		Outcomes Assessed
Final Written Examination	50%		2,3,4
Continuous Assessment	50%		
Practical		40%	1,6
Assignment		10%	5

Assessment Criteria

<40%: No understanding of the basic principles of microbial ecology and bioremediation.
40%-49%: The student has a basic understanding of the fundamental principles of the module. May have some difficulty with applying this theory.
50%-59%: All the above in addition to evaluation of practical data.
60%-69%: In addition to the above plus show a high level of competence and efficiency in the subject area.
70%-100%: Student will demonstrate a high level of understanding plus be able to critically evaluate and discuss the significance of the module.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	36	
Independent Learning	75	

Essential Material

- " Moodle e-learning platform." Lecturers may also assign reading from relevant current literature (e.g. journal articles).
- Madigan, M.T., J.M. Martinko, D.A. Stahl and D.P. Clark. _Brock Biology of Microorganisms_. 13th ed. UK: Benjamin Cummings (Pearson), 2011.

Supplementary Material

- Northup, D.E. and L.L. Barton. _Microbial Ecology_. USA: Wiley-Blackwell, 2011.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Biology Lab